Aerospace Manufacturing Core
Study Guide

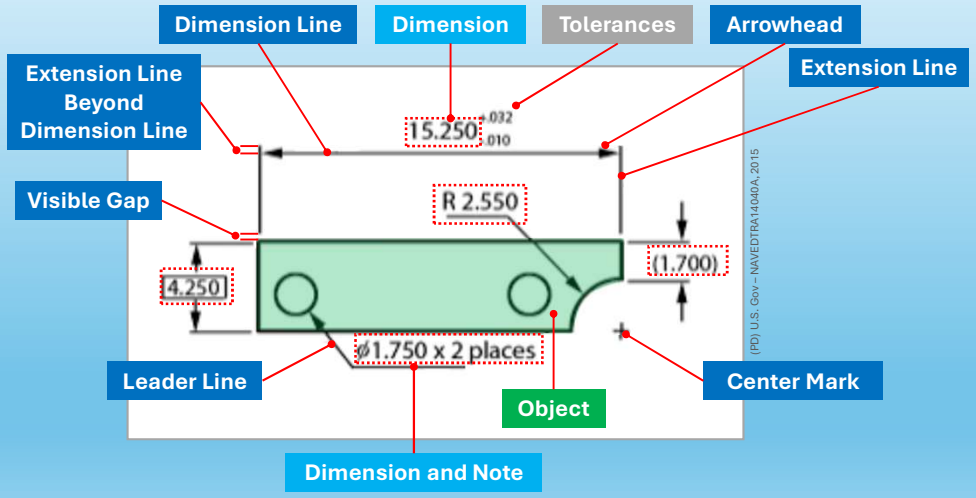
"AMCSG-304"

Blueprints & Documents

Dimensions & Tolerances

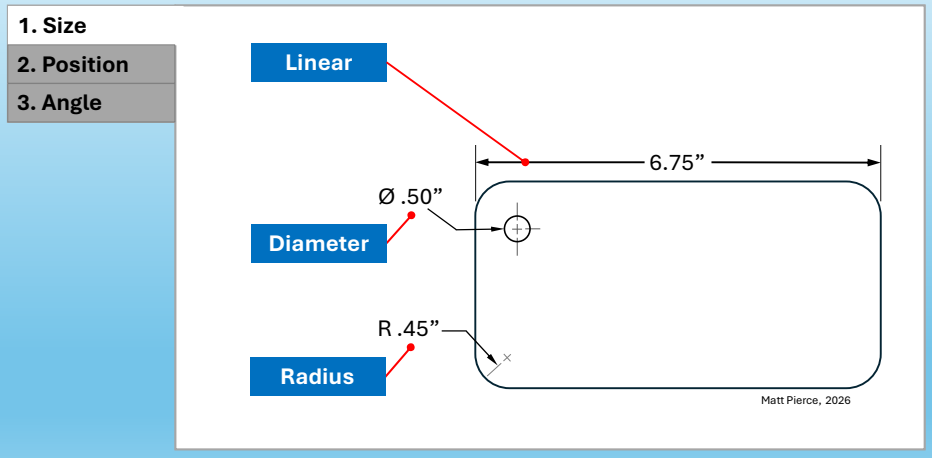
Dimensions

Dimensions: Overview



- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf

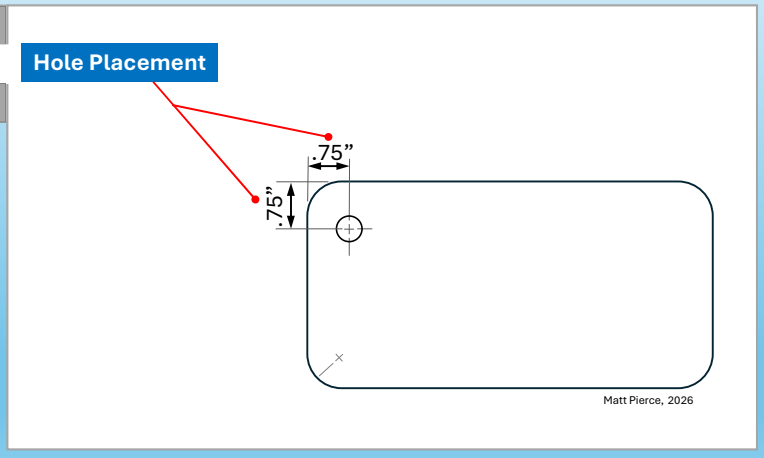
Dimension Types: “Size” (1 of 3)



- Google: ”describe types of dimensions on a blueprint: size, position, and angle. show examples.”

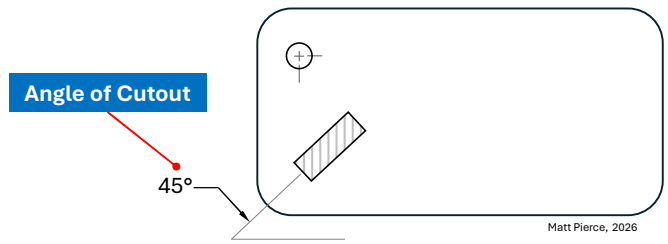
Dimension Types: “Position” (2 of 3)

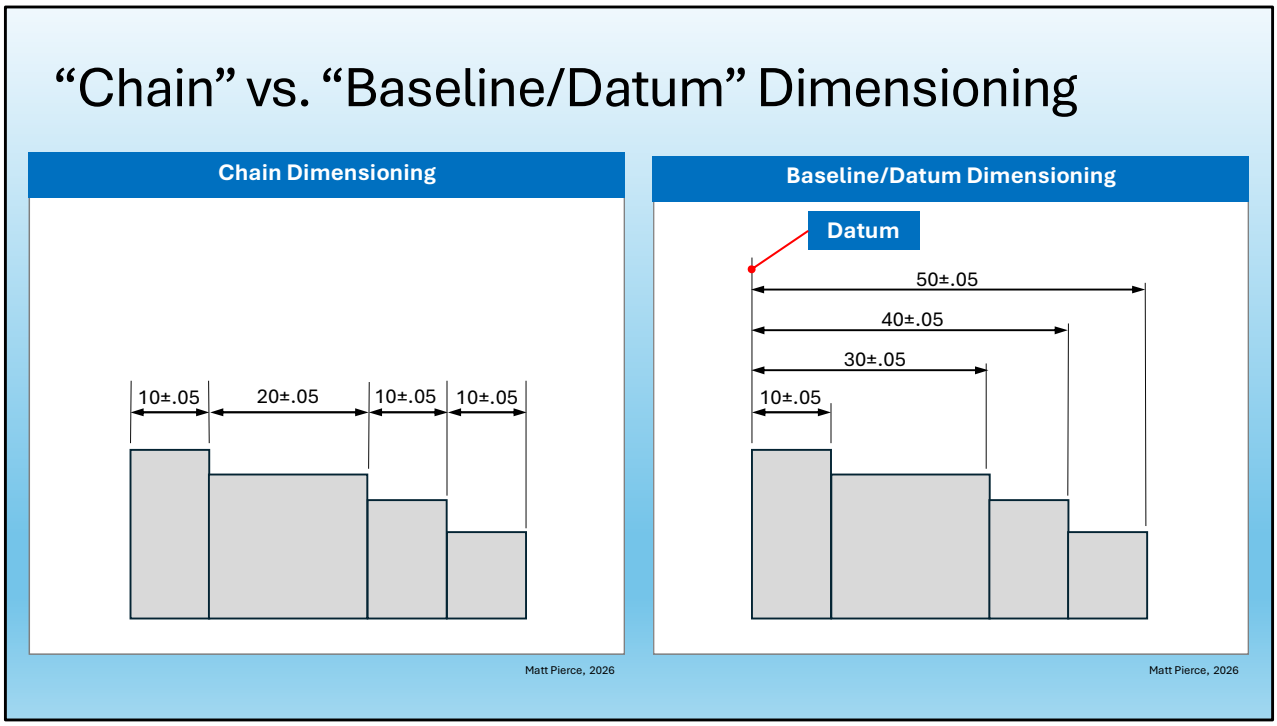
1. Size
2. Position
3. Angle



Dimension Types: “Angle” (3 of 3)

1. Size
2. Position
3. Angle





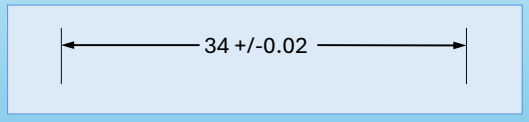
- ChatGPT: in a single diagram, show the difference between "chain dimensioning" and "datum dimensioning"

Tolerances

Definition of Tolerances

"Tolerances are the allowable variations in a measurement"
- Dr. Google

- **Example:** 34" +/- 0.02"

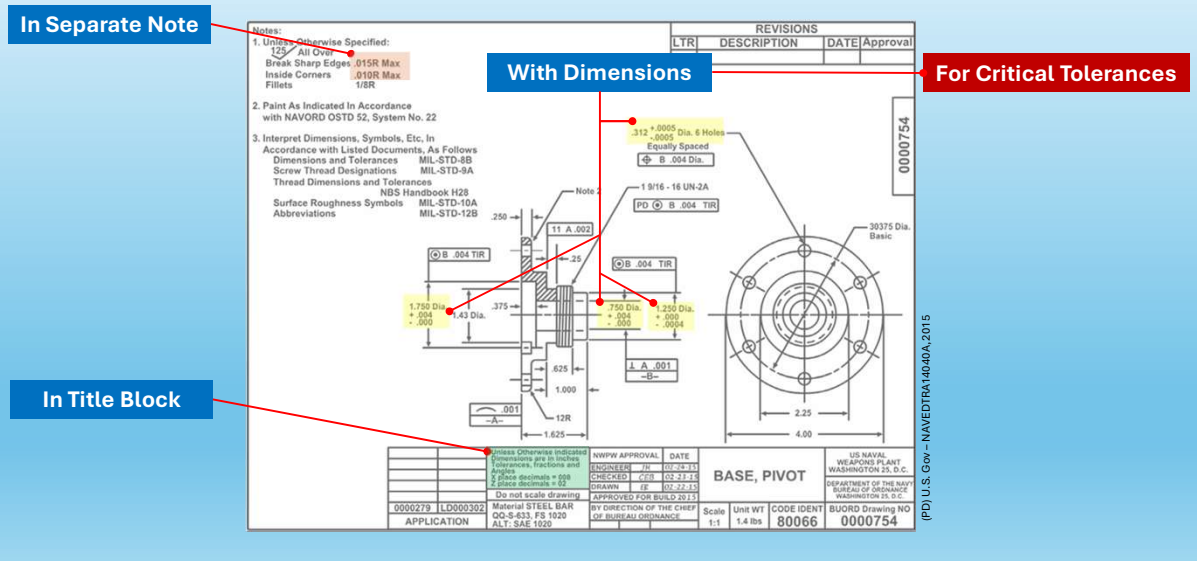


Matt Pierce, 2026

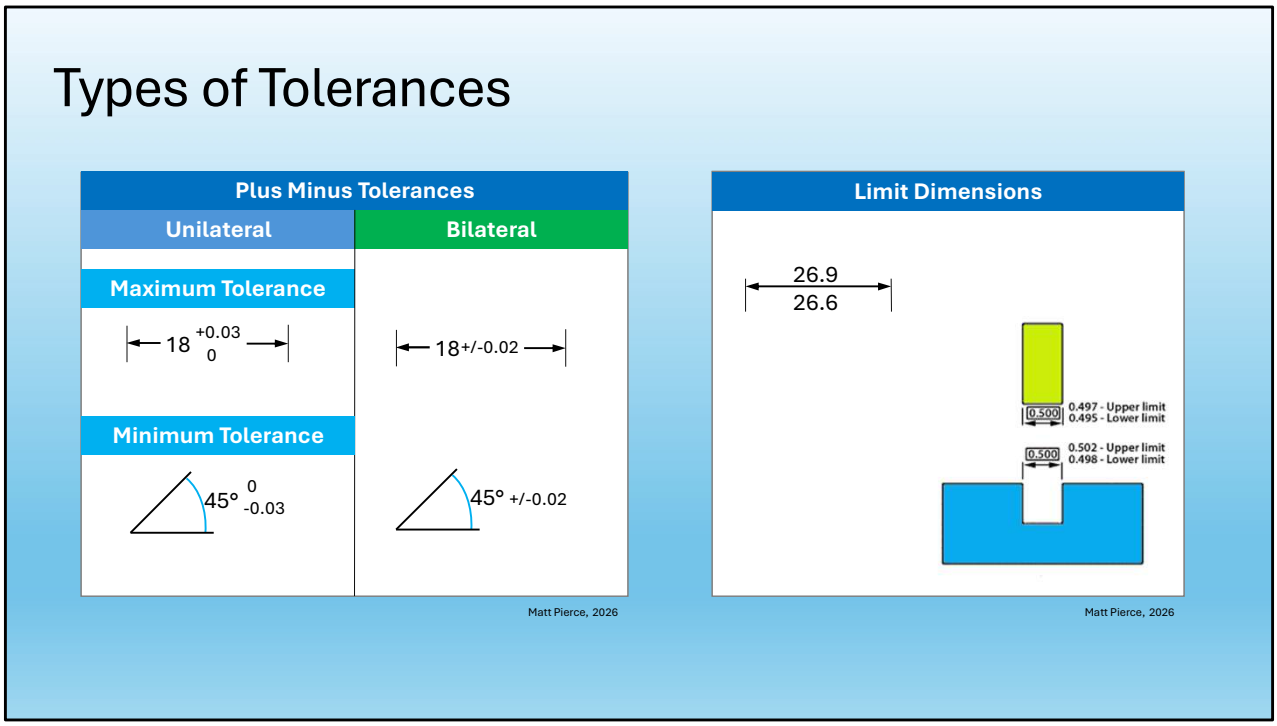
= 34 inches

with a tolerance of plus or minus two-one-hundredths of an inch

Location of Tolerances on Blueprint



- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf
(pg 69)



- ChatGPT: use diagrams to show the different types of tolerances on aerospace blueprints. In particular, "plus and minus tolerances", "limit dimensions", "maximum tolerances" and "minimum tolerances"

Q&A

Practice Q&A's (Q1-Q8)

Q1: What are numeric distance measurements on a blueprint called?

A: Dimensions

Q2: What is the type of line connecting a note to a drawing object (e.g.: a to a hole)?

A: Leader Line

Q3: What type of dimension shows linear length, or diameter, or radius?

A: Size Dimensions

Q4: What type of dimension shows where the center of a hole is relative to the corner of an object?

A: Position Dimensions

Q5: What type of dimension shows the angle of one element relative to another?

A: Angle Dimensions

Q6: What type of dimensioning shows relative size, one piece to another in a single set of connected lines?

A: Chain Dimensioning

Q7: What type of dimensioning shows absolute size from a single reference point or surface for each piece?

A: Baseline Dimensioning
(or Datum Dimensioning)

Q8: What are allowable variations in a measurement?

A: Tolerances

Practice Q&A's (Q9-Q14)

Q9: Y/N: Tolerances can be located in the Title Block?

A: Yes. Typical tolerances can be listed in the Title Block grid

Q10: Y/N: Tolerances can be located in the Notes?

A: Yes. Typical tolerances can also be listed with more detail in the Notes

Q11: Y/N: Tolerances can be listed on the drawing object?

A: Yes. The critical tolerances are called out specifically with dimension and extension lines in the drawing

Q12: What type of tolerance is "20 0 +0.03"?

A: Maximum Unilateral Tolerances

Q13: What type of tolerance is "20 -0.03 0"?

A: Minimum Unilateral Tolerances

Q14: What type of tolerance is "20 +/- 0.02"?

A: Bilateral Tolerances

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg